

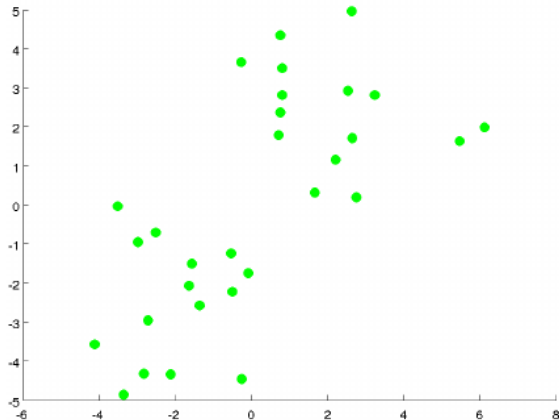
Kmeans Clustering

Assit.Prof.Dr.Emre Özbilge

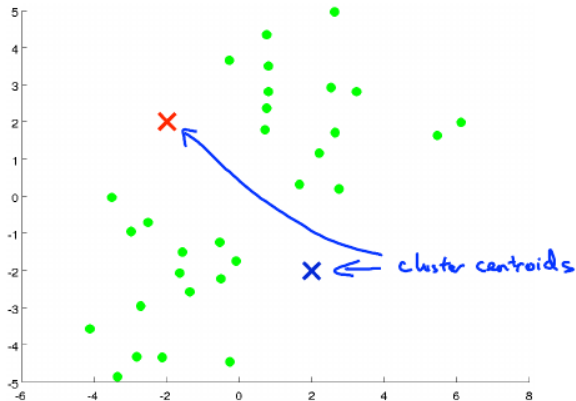
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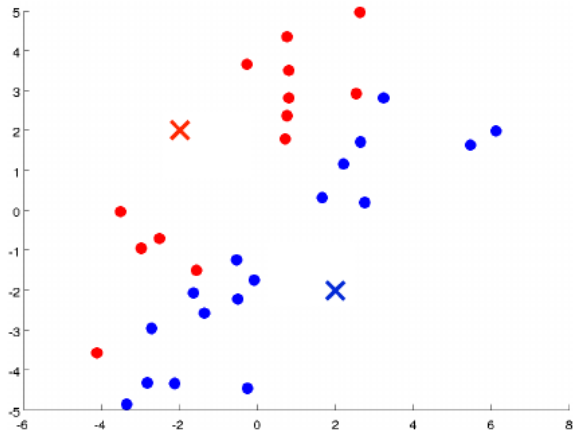
Clustering – K-means Algorithm I



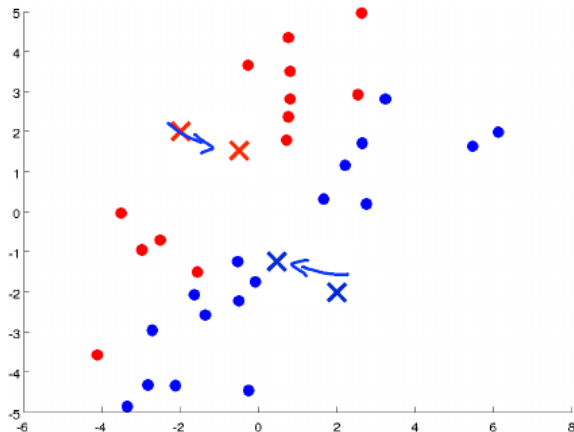
Clustering – K-means Algorithm II



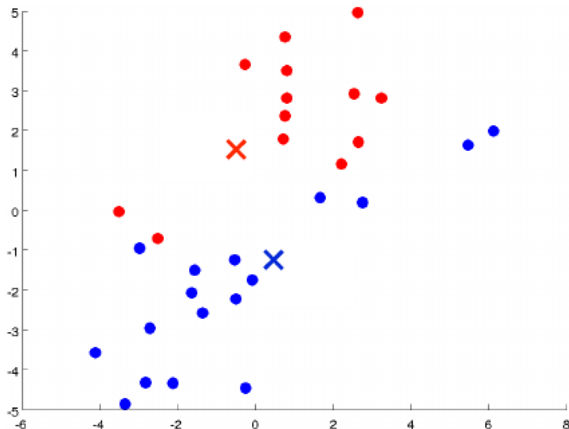
Clustering – K-means Algorithm III



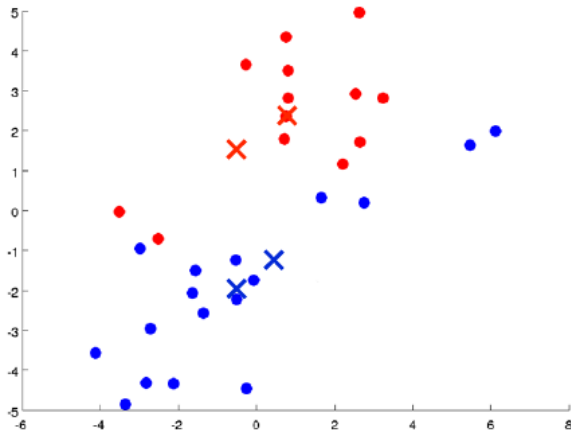
Clustering – K-means Algorithm IV



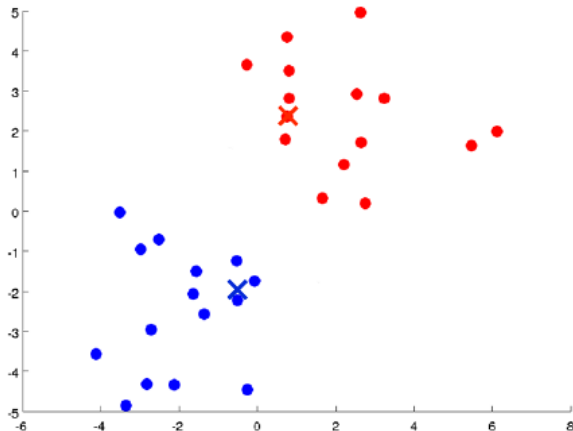
Clustering – K-means Algorithm V



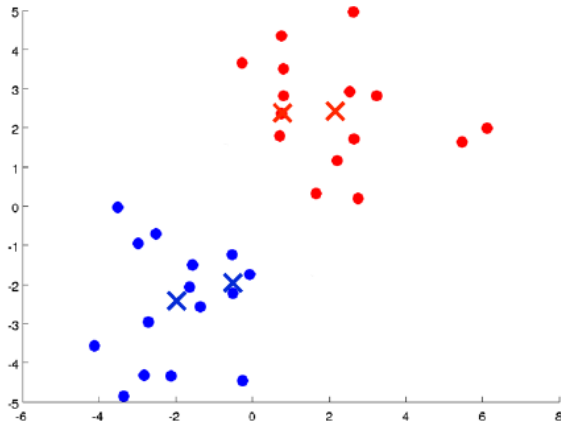
Clustering – K-means Algorithm VI



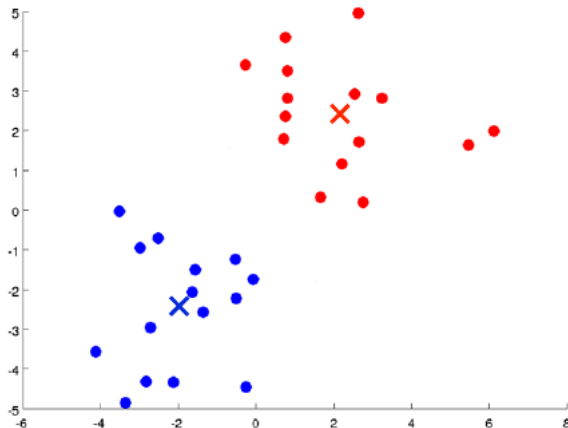
Clustering – K-means Algorithm VII



Clustering – K-means Algorithm VIII



Clustering – K-means Algorithm IX



Clustering – K-means Algorithm X

K-means algorithm

$$\mu_1 \quad \mu_2$$

Randomly initialize K cluster centroids $\mu_1, \mu_2, \dots, \mu_K \in \mathbb{R}^n$

Repeat {

Cluster assignment step

for $i = 1$ to m

$c^{(i)}$:= index (from 1 to K) of cluster centroid closest to $x^{(i)}$

$$\min_k \|x^{(i)} - \mu_k\|^2$$

Move centroid

for $k = 1$ to K

$\rightarrow \mu_k$:= average (mean) of points assigned to cluster k

$$x^{(1)}, x^{(5)}, x^{(6)}, x^{(10)}$$

$$\rightarrow c^{(1)}=2, c^{(5)}=2, c^{(6)}=2, c^{(10)}=2$$

$$\mu_2 = \frac{1}{4} [x^{(1)} + x^{(5)} + x^{(6)} + x^{(10)}] \in \mathbb{R}^n$$

$$\text{dist}(x^{(i)}, \mu_k) = \|x^{(i)} - \mu_k\|^2 = \sum_{j=1}^n (x_j^{(i)} - \mu_{k,j})^2$$

(1)

Cost Function

Cost function for K-means

$$J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K) = \frac{1}{m} \sum_{i=1}^m \|x^{(i)} - \mu_{c^{(i)}}\|^2$$

Repeat {

- # Assign points to cluster centroids

for $i = 1$ to m :

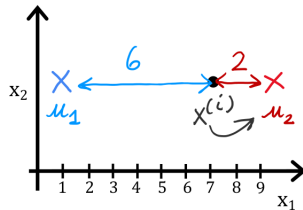
- $c^{(i)}$ = index of cluster centroid closest to $x^{(i)}$

- # Move cluster centroids

for $i = 1$ to K :

- μ_k = average of points in cluster

}



Selecting Optimal Cluster Number

Choosing the value of K

Elbow method

